

JIHWAN KIM

STRATEGY & ANALYTICS | DATA-DRIVEN OPERATIONS | SUSTAINABILITY

Tokyo, Japan | tho3ov@gmail.com | Korean (native), English & Japanese (professional working proficiency)

PROFESSIONAL SUMMARY

Ph.D. oceanographer and analytics professional translating scientific, geospatial, and operational data into commercial and regulatory decisions. Experienced in predictive modeling, digital-product development, compliance analytics, and international stakeholder management across France, Japan, and Asia-Pacific markets. Seeking strategy and transformation roles focused on data-driven growth and operational improvement.

Core capabilities: Strategy & structured problem solving | Predictive modeling & scenario analysis | Data and digital products | Operations & compliance analytics | International stakeholder management

PROFESSIONAL EXPERIENCE

Ocean Engineer, Sustainable Fisheries Management

2025–Present

Team

Collecte Localisation Satellites (CLS) | France & Japan

- Deliver satellite-based fisheries intelligence and fishing-area recommendations to **10+ commercial clients** and **approximately 150 tuna vessels** across Korea, Japan, Taiwan, the Philippines, and the United States, covering operations in the Pacific, Indian, and Atlantic Oceans.
- Designed and **automated daily Favorability Maps** integrating sea-surface temperature, chlorophyll, currents, fronts, weather, and species-environment relationships to continuously rank potential fishing areas for voyage and search planning.
- Developed and validated climate-informed statistical and machine-learning models whose outputs closely track purse-seine catch variability; produce **2-3-month-ahead potential fishing-ground maps** and statistically predict ENSO/IOD-driven catch shifts **up to six months in advance**.
- Contribute to the design and implementation of CATSAT, a satellite-based fisheries intelligence platform, by translating operational needs and predictive models into data requirements, algorithms, and decision-support features.
- Partner with client-facing, scientific, and technical teams in France and Japan to convert vessel-operating needs into analytical workflows, product requirements, and actionable outputs.

Data & Fisheries Science Advisor (Consultant)

2024–2026

North Pacific Fisheries Commission (NPFC) | Tokyo, Japan

- Advise an intergovernmental fisheries commission on resource management, compliance monitoring, and data governance, presenting evidence-based findings and recommendations to representatives from **7+ member delegations**.
- Processed and quality-controlled **approximately 9 million VMS records** across three years (approximately 3 million annually) and designed matching algorithms linking VMS, AIS, transshipment declarations, boarding-inspection records, and other operational datasets, identifying **100+ confirmed data inconsistencies and anomalous cases per year**.
- Developed GIS-based tools to visualize catch-and-effort patterns and support cross-country analysis of North Pacific fisheries, including Pacific saury, chub mackerel, and neon flying squid.
- Produced and presented Fisheries, Transshipment, and VMS overviews at NPFC technical meetings and compiled commission data for the FAO Fisheries and Resources Monitoring System.
- Applied stochastic simulation, statistical forecasting, and deep-learning super-resolution methods to investigate climate-driven changes in fish distribution and strengthen evidence-based resource-management discussions.

Postdoctoral Fellow

2023–2024

North Pacific Fisheries Commission & Tokyo University of Marine
Science and Technology | Tokyo, Japan

- Led quantitative studies connecting basin-scale climate variability and Kuroshio dynamics with changes in Pacific saury and chub mackerel distribution, abundance, and catch performance.
- Developed monitoring concepts and algorithms for IUU-fishing detection by integrating vessel-tracking and air-surveillance information, and proposed measures to improve catch-reporting accuracy.
- Designed and implemented geospatial data products that enabled commission members to compare fisheries activity consistently across jurisdictions and reporting formats.

EARLIER EXPERIENCE & SELECTED PROJECTS

Earth Science Teacher

2018–2019

Sehwa High School | Seoul, Republic of Korea

Researcher – Korea/Tsushima Strait Observation System

2020–2023

Seoul National University | Republic of Korea & Japan

- Designed a replacement observation solution, new data-analysis method, and cloud data server to restore submarine-cable voltage monitoring after equipment failure, enabling continued estimation of volume transport through the Korea/Tsushima Strait.

EDUCATION

Ph.D. in Physical Oceanography

2019–2023

Seoul National University, School of Earth and Environmental Sciences | Seoul, Republic of Korea

- Focus: climate–fisheries forecasting, upper-ocean circulation, machine learning, and geospatial analysis.

B.S. in Earth Science Education

2013–2018

Seoul National University, College of Education | Seoul, Republic of Korea

REPRESENTATIVE MODELING & ADVISORY ENGAGEMENTS

- Developed stochastic spatio-temporal forecasting and a **multi-head attention U-Net** to improve seasonal fishing-ground outlooks and generate physically consistent super-resolution distributions from coarse data.
- Linked ENSO/IOD, Kuroshio, and Pacific Decadal Oscillation variability to tuna and mackerel catch patterns several months in advance, translating climate signals into operational and management insights.

TECHNICAL & BUSINESS SKILLS

Analytics & technology: Statistical forecasting, regression, scenario analysis, spatio-temporal modeling, machine learning, deep learning, geospatial analysis; Python, MATLAB, SQL/MySQL, GIS, dashboards, and data visualization

Additional tools: Working knowledge of Julia and Fortran; web-application prototyping, database design, and requirements definition

Consulting-relevant strengths: Problem structuring, evidence synthesis, executive-ready reporting, requirements definition, cross-functional collaboration, international stakeholder communication

SELECTED PUBLICATIONS & RECOGNITION

- Published peer-reviewed research in **Scientific Reports and Frontiers in Marine Science** and presented fisheries forecasting, vessel-monitoring, and climate–ecosystem findings to NPFC committees, PICES, AGU, and the Korea Oceanographic Society.
- **2nd Prize**, JOISS Marine Science Big Data Competition (2021 and 2022; UN Ocean Decade Activity); Succeeding Generations of Fundamental Sciences Fellowship, Seoul National University (full scholarship, 2019–2020).